24/2 (23)

No of additional sheets

LOURDES MATHA COLLEGE OF SCIENCE & TECHNOLOGY
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ANSWER BOOK

Name Adwaith. A.M Class No. 05 Branch MCA

Subject ~~Machine learning~~ Data science Semester S3

Mark Tabulation

	Q.1	Q.2	Q.3	Q.4	Q.5	Q.6	Q.7	Q.8	Q.9	Q.10	Q.11	Q.12	Q.13	Q.14
Mark Divisions	3	1½	3	2	3	2	4½		a.1/e b.2½					
Total														
Grand Total														23

ordinary least savuare method is a linea
regression technique used to find the
line for a set of data points. It is a
method because it is easy to use and
decent results. The ordinary least savua
can be defined as linear regression
that is used to estimate the unkno

In mathematical terms, th

minimize $\varepsilon (y_i - \bar{y_i})^2$

where: y_i is the actual value and $\bar{y}_i$ is the value.

$\bar{y} = mx + b$

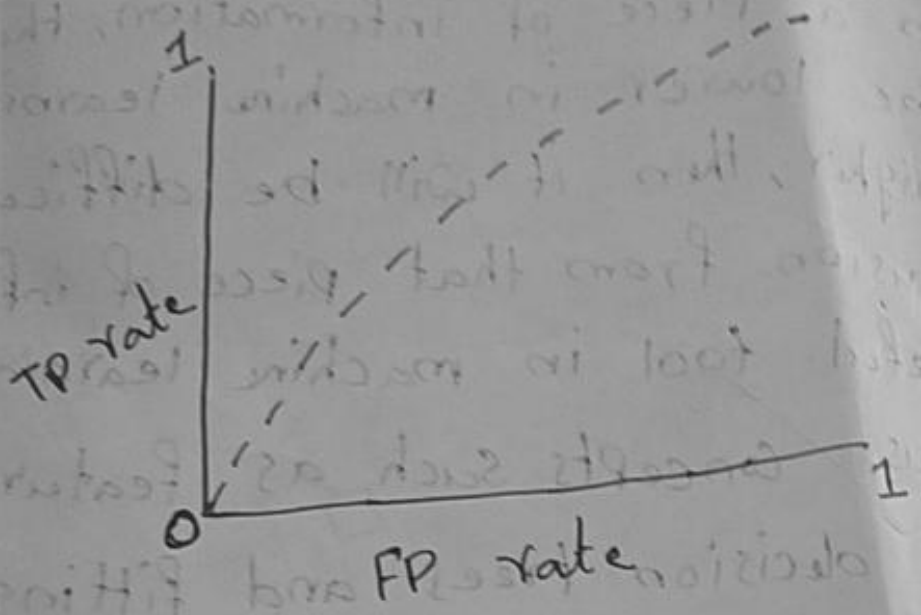
where b is the intercept and m is the sl
the line. The coefficient m and b can also
as coefficients of determination. ordinary
squaare method finds the best fit line for
by minimizing the sum of squared erro
residuals between actual and predicted

An ROC curve is the graph showing the
of a classification model at all classificati
The curve plots two parameters,

* True Positive rate (TPR)
* False Positive rate (FPR)

For recall and

FPR = $\frac{FP}{FP+TN}$



Typical ROC C

3 An ROC curve plots TPR vs FPR at differe
classification thresholds. Lowering the classific
threshold classifies more item as positive
increasing both false positives and true p

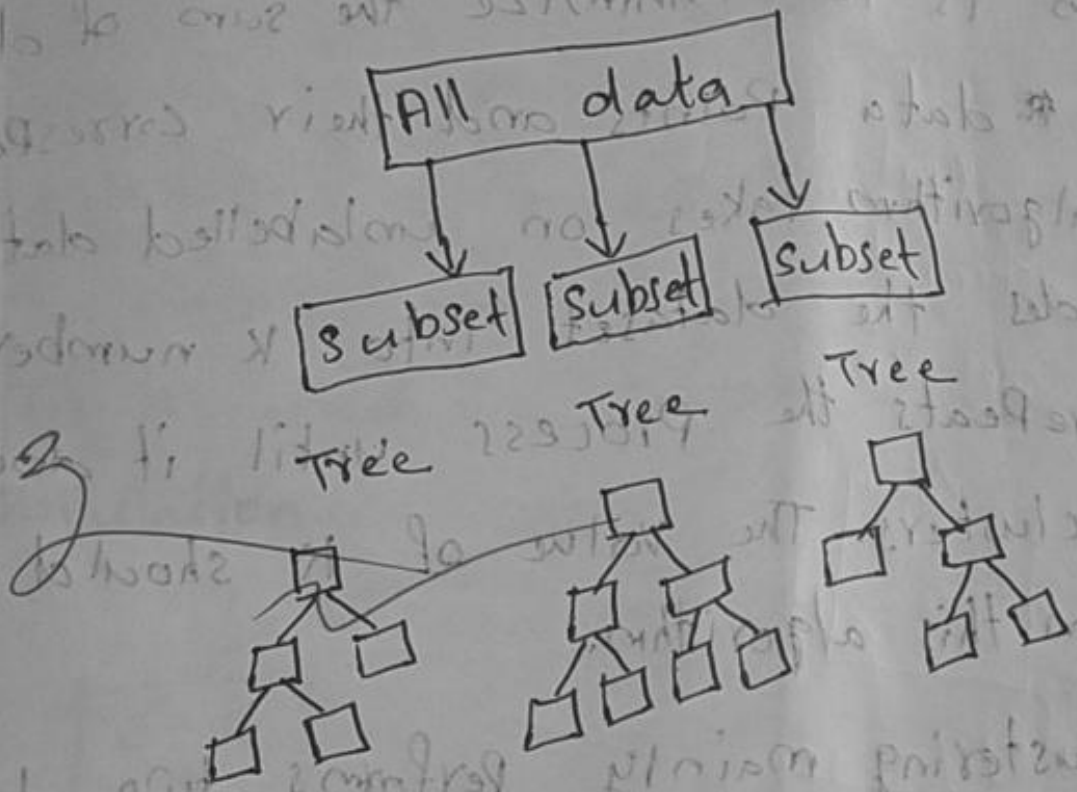
5, Entropy is defined as the randomness or
the disorder of the information being Pro
machine learning. Further in other words
that entropy is the machine learning
measures the unpredictability or impur
system

when information is pro
every piece of information has a spe
to make and can be used to draw
from it. so if it is easier to draw a
conclusion from a piece of information,
entropy will be lower in machine lea
if entropy is higher, then it will be dif
draw any conclusion from that piece of
Entropy is a useful tool in machine le
understand various concepts such as fe
selection, building decision trees and f
classification models etc.

6. A Random forest is a supervised
that uses an ensemble learning meth
of a multitude of decision trees, th
which is the consensus of the best a
problem. Random forest can be used
or regression. It builds decision
different samples and take their m
for classification and average in c

If performs better results for classification problems:

ADDITIONAL SHEET



K-means clustering is an unsupervised lea
algorithm that is used to solve the c
problems in machine learning or data
groups the unlabeled dataset into di
Here K defines the number of pre-defin
that need to be created in the p
K=2, there will be two clusters an
will be three clusters and so on.
cluster the dat

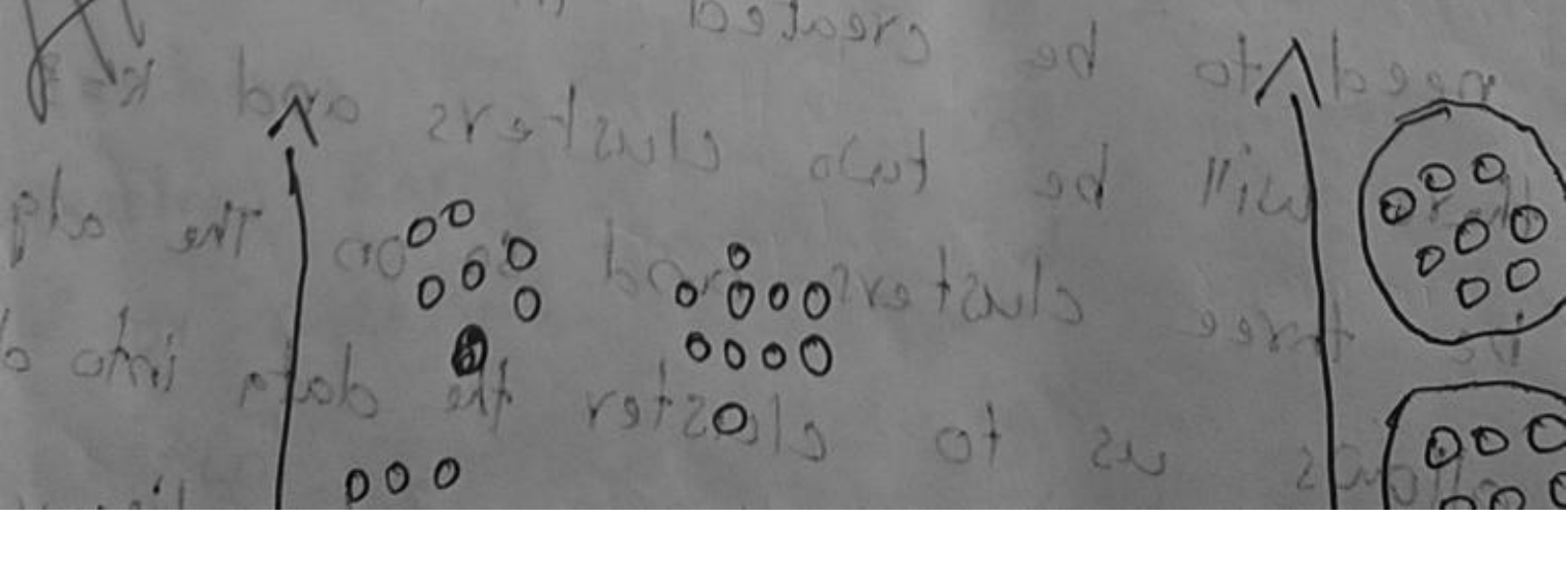
It is centroid based algorithm, each
is associated with a centroid. The main aim
this algorithm is to minimize the sum of
between the data points and their corre
clusters. The algorithm takes an unlabelled
as input, divides the dataset into k nu
clusters and repeats the process until it
find the best clusters. The value of k sho
predetermined in this algorithm.

The K-means clustering mainly performs tw

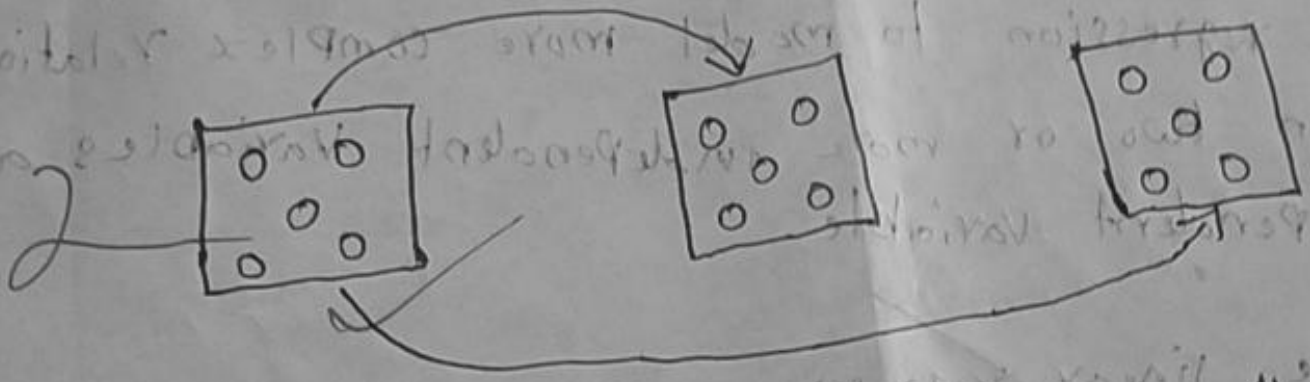
* Determines the best value of K center
centroids by an iterative process.

* Assigns each dataPoint to its closes

Those data points which are near to
particular K-center, create a clust



4, Bootstrap sampling is the method in which we take
a simple data repeatedly with replacement fro
a dataset to estimate a population parameter.
It is used to determine various parameters of
population.



2 A common strategy is to grow the tree
each node contains a small number of in
then use pruning to remove nodes that
provide additional information. Pruning
reduce the size of a learning tree
reducing predictive accuracy as m
a cross-validation set. There are many
that differ in the m

It is ...

9, a, correlation

correlation measures the relationship between variables.

b, multiple linear regression

It is a statistical technique that uses linear regression to model more complex between two or more independent Vari one dependent variable.

c, simple linear regression

simple linear regression is used to best relationship between a single and output variable provided that are continuous in nature.